Bubble Sort and Finding the First and Last Occurrence of a Target Element

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1. Problem Statement

Write a C program that reads the size and the elements of a variable-length integer array nums[]. The array may contain duplicate elements. The array is passed by address to a function Bubble_sort(), which sorts it in ascending order using the Bubble Sort technique. The sorted array is then passed to another function occurrence(), which reads an integer target and finds its first and last occurrences in the sorted array. Pointers must be used to pass the array between the functions.

2. Objectives

- To read an array whose size is decided by the user at run time (dynamic memory allocation).
- To pass an array to functions using pointers.
- To sort an array in ascending order using the Bubble Sort technique.
- To find the first and last index of a target value in a sorted array containing duplicates.
- To display -1, -1 when the target is not present.

3. Concepts Used

Pointers: the array is passed to both functions as an int * (address of the first element) and accessed using pointer notation *(arr + i).

Dynamic memory allocation: malloc() creates the array according to the size entered by the user, and free() releases it at the end.

Bubble Sort: repeatedly compares adjacent elements and swaps them if they are in the wrong order. After each pass the largest remaining element moves to its correct position at the end.

Linear search for first and last occurrence: the array is scanned once. The first match index is stored only once, while the last match index keeps updating on every match.

4. Algorithm

4.1 Main Function

1. Start.
2. Read the size n of the array and allocate memory for n integers.
3. Read the n elements into nums[].
4. Call Bubble_sort(nums, n) to sort the array.
5. Display the sorted array.
6. Call occurrence(nums, n).

7. Free the allocated memory and stop.

4.2 Bubble_sort(arr, n)

1. Repeat for i = 0 to n - 2:
2. For j = 0 to n - i - 2, compare arr[j] and arr[j + 1].
3. If arr[j] > arr[j + 1], swap them.
4. If no swap happened in a pass, the array is sorted, so stop early.

4.3 occurrence(arr, n)

1. Read the integer target. Set first = -1 and last = -1.
2. For i = 0 to n - 1: if arr[i] equals target, then set first = i (only if first is still -1) and set last = i.
3. If first is still -1, print "-1, -1".
4. Otherwise print the value and index of the first and last occurrences.

5. Source Code

```
/*
 * Program : Bubble Sort and First/Last Occurrence of a Target
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 */

#include <stdio.h>
#include <stdlib.h>

/* Function prototypes */
void Bubble sort(int *arr, int n);
void occurrence(int *arr, int n);

int main(void)
{
    int n, i;
    int *nums;

    /* Read the size of the array */
    printf("Enter the size of the array: ");
    if (scanf("%d", &n) != 1 || n <= 0)
    {
        printf("Invalid size.\n");
        return 1;
    }

    /* Allocate memory for a variable-length array */
    nums = (int *)malloc(n * sizeof(int));
    if (nums == NULL)
    {
        printf("Memory allocation failed.\n");
        return 1;
    }

    /* Read the elements */
    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++)
    {
        scanf("%d", &nums[i]);
    }

    /* Sort the array (passed by address) */
    Bubble_sort(nums, n);

    printf("Sorted array: ");
    for (i = 0; i < n; i++)
    {
        printf("%d ", *(nums + i));
    }
    printf("\n");

    /* Search for the target in the sorted array */
    occurrence(nums, n);

    free(nums);
    return 0;
}

/* Sorts the array in ascending order using Bubble Sort */
void Bubble_sort(int *arr, int n)
{
    int i, j, temp, swapped;

    for (i = 0; i < n - 1; i++)
    {
        swapped = 0;
        for (j = 0; j < n - i - 1; j++)
        {
            if (*(arr + j) > *(arr + j + 1))
            {
                temp = *(arr + j);
                *(arr + j) = *(arr + j + 1);
                *(arr + j + 1) = temp;
                swapped = 1;
```

```
            }
        }
        if (swapped == 0)   /* already sorted, stop early */
        {
            break;
        }
    }
}

/* Reads a target and finds its first and last occurrence */
void occurrence(int *arr, int n)
{
    int target, i;
    int first = -1, last = -1;

    printf("Enter the target: ");
    scanf("%d", &target);

    for (i = 0; i < n; i++)
    {
        if (*(arr + i) == target)
        {
            if (first == -1)
            {
                first = i;      /* first match */
            }
            last = i;           /* keeps updating to the last match */
        }
    }

    if (first == -1)
    {
        printf("-1, -1\n");
    }
    else
    {
        printf("First occurrence: %d at index %d\n", *(arr + first), first);
        printf("Last occurrence: %d at index %d\n", *(arr + last), last);
    }
}
```

6. Sample Input and Output

The program was compiled with gcc (using the -Wall -Wextra options, with no warnings) and tested with the three examples given in the problem statement. The results match the expected outputs.

Test Case 1: Target present with duplicates

```
Enter the size of the array: 6
Enter 6 elements: 5 7 7 8 8 10
Sorted array: 5 7 7 8 8 10
Enter the target: 8
First occurrence: 8 at index 3
Last occurrence: 8 at index 4
```

Test Case 2: Target present only once

```
Enter the size of the array: 5
Enter 5 elements: 5 7 7 8 10
Sorted array: 5 7 7 8 10
Enter the target: 8
First occurrence: 8 at index 3
Last occurrence: 8 at index 3
```

Test Case 3: Target not present

```
Enter the size of the array: 6
Enter 6 elements: 5 7 7 8 8 10
Sorted array: 5 7 7 8 8 10
Enter the target: 6
-1, -1
```

Test Case 4: Unsorted input

```
Enter the size of the array: 7
Enter 7 elements: 10 8 5 7 8 7 1
Sorted array: 1 5 7 7 8 8 10
Enter the target: 7
First occurrence: 7 at index 2
Last occurrence: 7 at index 3
```

7. Working Explanation

Consider Test Case 1 with the array 5 7 7 8 8 10 and target 8. The array is already in ascending order, so Bubble_sort() makes no swaps in its first pass and stops early. In occurrence(), the loop scans the array from index 0. The first 8 is found at index 3, so first = 3 and last = 3. The next 8 is found at index 4, so last is updated to 4. No further matches exist, so the program prints the first occurrence at index 3 and the last occurrence at index 4.

In Test Case 3 the target 6 is never found, so first stays at -1 and the program prints -1, -1.

8. Complexity

- Bubble_sort(): worst and average case time $O(n^2)$; best case $O(n)$ for an already sorted array because of the early-exit check.
- occurrence(): time $O(n)$ for a single scan of the array.

- Extra space: O(1), because sorting is done in place using pointers.

9. Conclusion

The program successfully reads an array of any size, sorts it in ascending order using Bubble Sort through pointer-based function calls, and locates the first and last occurrences of a target element, handling duplicates and the not-found case correctly. This project helped in understanding pointers, dynamic memory allocation, array passing between functions, and the working of the Bubble Sort algorithm.